

Fine-Grained Emotion Detection in Text using DistilBERT

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Course: CSE-6363-003 Machine Learning

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Abstract

Traditional sentiment analysis systems often restrict emotional interpretation to positive, negative, or neutral sentiments, overlooking the nuanced spectrum of human emotions. In this project, we developed a fine-grained, multi-label emotion classification system leveraging the Go Emotions dataset and a fine-tuned Distil BERT Transformer model. Our approach addressed challenges such as class imbalance, multi-emotion co-occurrence, and fairness across emotion categories. By implementing class-balanced loss, dual learning rates, and per-emotion threshold tuning, we achieved a strong overall Micro-F1 score of 0.8339. Our system demonstrates that modern NLP techniques can successfully model the complexity of human affective expression.

Datasets: <https://www.kaggle.com/datasets/debarshichanda/goemotions/data>

1. Introduction

Understanding human emotion through language is one of the most complex and fascinating challenges in Natural Language Processing (NLP). Emotions play a crucial role in communication, influencing how humans perceive, interpret, and respond to text. Traditional sentiment analysis techniques focus primarily on identifying the polarity of text—classifying it as positive, negative, or neutral. While effective for basic opinion mining, these models fail to capture the subtlety and overlap of emotional expression found in natural human discourse. For instance, a single statement such as “*I’m nervous but excited for my first day at work*” simultaneously conveys *fear* and *joy*. Recognizing such nuanced emotional layers requires models capable of understanding multi-dimensional affective states rather than binary sentiment categories.

Recent advances in deep learning, particularly with Transformer-based architectures, have transformed NLP research. Models like BERT and Distil BERT can capture contextual semantics with remarkable precision by analyzing the relationships between words within entire sentences rather than sequentially. This context-awareness makes them ideal for emotion recognition tasks where meaning is highly dependent on phrasing, tone, and context. However, fine-grained emotion detection remains a challenging problem due to issues like class imbalance, multi-label dependencies, and context ambiguity—where the same word or phrase may convey different emotions depending on context. Addressing these challenges requires careful dataset design, model tuning, and fairness-oriented training approaches.

This project aims to design a fine-grained emotion detection system capable of recognizing multiple concurrent emotions within textual input. Using the Go Emotions dataset, a comprehensive corpus of Reddit comments annotated for 27 emotions, we consolidated the labels into 8 broad emotion categories and fine-tuned the Distil BERT model for multi-label classification. Our focus was not only on achieving high accuracy but also on improving fairness and interpretability across underrepresented emotions such as *fear* and *disgust*. The resulting model demonstrates strong generalization, achieving a Micro-F1 score of 0.8339, validating that smaller, efficient Transformer models can rival larger ones while maintaining balanced performance and computational efficiency.

2. Dataset Description

The dataset used for this project is the Go Emotions dataset, introduced by Demszky et al. (2020) at Google Research. It is one of the most comprehensive and fine-grained emotion-annotated corpora currently available for natural language processing tasks. The dataset consists of approximately 58,000 Reddit comments manually labeled for 27 distinct emotion categories along with a neutral tag. The emotion taxonomy in Go Emotions was derived from psychological theories of affect, encompassing a wide

range of human emotions such as *joy*, *sadness*, *admiration*, *amusement*, *anger*, *fear*, *surprise*, and *disgust*. Each comment may contain multiple emotion labels, reflecting the fact that human emotions are rarely singular or isolated. For this project, the 27 fine-grained emotion classes were consolidated into eight broader categories—*joy*, *admiration*, *surprise*, *anger*, *sadness*, *disgust*, *fear*, and *neutral*—to reduce sparsity and improve the model’s ability to generalize across semantically similar emotions. This mapping preserved the expressiveness of the dataset while simplifying the classification task to a manageable level for the fine-tuning process.

An exploratory analysis of the dataset revealed a highly imbalanced distribution of emotion classes, where certain categories such as *joy* and *admiration* were significantly more frequent than others like *fear* or *disgust*. This imbalance poses a common challenge in emotion recognition tasks, often leading models to become biased toward dominant emotions. To mitigate this issue, we implemented several pre-processing and data-handling strategies, including multi-label stratified splitting to ensure proportional representation of all emotion classes across training, validation, and test sets. Additionally, we applied text normalization and cleaning to remove noise such as hyperlinks, emojis, and non-linguistic symbols, thereby improving the quality of input text. Tokenization was performed using Distil Bert’s Word Piece tokenizer, converting text into subword units that help the model handle rare or unseen words effectively. These steps ensured that the data fed into the model was balanced, clean, and representative of the wide range of emotional diversity present in online communication.

3. Data Preparation and Exploration

1. Distribution Analysis: The initial distribution analysis of the Go Emotions dataset revealed a significant imbalance among emotion categories, posing one of the primary challenges in this project. Emotions such as *joy* and *admiration* appeared far more frequently, while others like *fear* and *disgust* were underrepresented. This uneven class distribution created the risk of model bias, where dominant emotions could disproportionately influence the learning process, leading to skewed or less reliable predictions for rare categories. Since the dataset follows a multi-label structure allowing multiple emotions to co-occur within a single text instance the imbalance became even more complex. Some combinations of emotions were scarcely represented, reducing the model’s exposure to diverse emotional contexts. To counter this, we implemented multi-label stratified splitting, ensuring that all subsets of the data (training, validation, and test) maintained a consistent distribution of emotion labels. This approach helped

preserve the diversity of emotional expressions while improving the fairness and generalizability of the final model.

2. Preprocessing: Preprocessing was a critical stage to ensure the dataset’s integrity and the reliability of subsequent model training. Since the GoEmotions dataset consists of user-generated Reddit comments, the raw data contained noise such as incomplete sentences, duplicate entries, URLs, emojis, and irrelevant symbols. These artifacts were removed systematically to retain only clean, meaningful textual content that accurately conveyed emotional signals. Additionally, we filtered out samples without emotion labels to avoid introducing ambiguity into the training set. After cleaning, all text was normalized, including lowercasing words and removing excessive punctuation and whitespace, to create consistency in linguistic structure. This process not only improved the model’s ability to detect emotional patterns but also ensured that each input represented genuine emotional expression rather than noise or formatting inconsistencies. Through these preprocessing steps, the dataset achieved a higher signal-to-noise ratio, forming a solid foundation for effective model fine-tuning.

3. Tokenization: The final step in the data preparation pipeline was tokenization, which converts textual input into a numerical format that the model can process. We utilized Distil Bert’s Word Piece tokenizer, a sub word-based tokenization technique that breaks words into smaller, meaningful units called “word pieces.” This approach is particularly effective for handling rare, compound, or unseen words that are common in user-generated content. For example, a word like *heartbreaking* may be split into *heart* and *breaking*, enabling the model to infer meaning from familiar subcomponents. Each token was mapped to a corresponding numerical ID from the distil BERT vocabulary, preserving the semantic relationships within the text. To ensure computational efficiency and uniformity, all tokenized sequences were standardized using padding and truncation to a fixed maximum length. This consistency allowed the model to process inputs in parallel without losing important contextual information. Tokenization thus transformed the cleaned dataset into structured, model-ready input while maintaining the emotional and linguistic nuances crucial for accurate classification.

3. Project Description

3.1 Description

The primary objective of this project was to develop a fine-grained, multi-label emotion detection system capable of identifying multiple emotions coexisting in a single text input. Traditional sentiment analysis models tend to classify

sentences into a single polarity—positive, negative, or neutral—oversimplifying the richness of human emotional expression. Our approach, by contrast, was designed to recognize that emotional states often overlap and interact. For example, the statement “*I can’t believe I did it—I’m scared but thrilled*” expresses both *fear* and *joy* simultaneously. To effectively model this complexity, we formulated the task as a multi-label classification problem, where the model assigns multiple emotion labels to each input. This framing allows for a more realistic interpretation of emotional language and improves the relevance of emotion recognition in real-world applications such as mental health analysis, customer feedback, and conversational AI systems.

At the core of our system is Distil BERT, a distilled version of BERT that preserves much of BERT’s linguistic understanding while being 40% smaller and 60% faster. Distil BERT operates on the Transformer architecture, leveraging self-attention mechanisms to capture long-range contextual relationships between words. This capability is particularly valuable for emotion detection, where meaning depends heavily on subtle shifts in phrasing and context. We fine-tuned Distil BERT by adding a custom classification head consisting of a fully connected layer with sigmoid activation, which outputs independent probabilities for each emotion class. This enables the model to make multiple predictions simultaneously rather than forcing a single-choice outcome. To prevent overfitting and enhance generalization, we incorporated dropout layers and used binary cross-entropy loss adapted for multi-label classification.

During training, we employed a combination of advanced optimization techniques to achieve balanced and stable performance. The AdamW optimizer was used with dual learning rates—a lower rate for the pre-trained layers of Distil BERT and a higher rate for the new classification head—to ensure smooth adaptation without catastrophic forgetting. We also introduced class-balanced loss weighting to address data imbalance, ensuring that rare emotions like *fear* and *disgust* were not overshadowed by frequent ones such as *joy*. Additionally, per-emotion threshold tuning was performed after training to optimize the trade-off between precision and recall for each emotion class individually. Collectively, these design choices led to a system that achieved both high predictive accuracy (Micro-F1 score: 0.8339) and fairness across all emotion categories, confirming the robustness of our approach.

3.2 Main References Used

1. Demszky, D. et al. *Go Emotions: A Dataset of Fine-Grained Emotions*. Google Research, 2020.
2. Sanh, V. et al. *Distil BERT: A Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter*. Hugging Face, 2019.
3. Devlin, J. et al. *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. NAACL, 2019.

3.3 Difference in Approach/Method

Previous research in emotion detection has largely relied on either traditional machine learning or early deep learning techniques, which often required extensive manual feature engineering and lacked the ability to capture nuanced contextual relationships within text. For instance, studies such as Alm, Roth, and Sproat (2005) employed classical algorithms like Naïve Bayes and Support Vector Machines (SVMs) using handcrafted linguistic features such as part-of-speech tags and sentiment lexicons. While these models performed reasonably well for simple sentiment tasks, they struggled to handle the ambiguity and overlapping nature of real human emotions. Later, Abdul-Mageed and Ungar (2017) introduced EmoNet, a Gated Recurrent Neural Network (GRNN)-based model that leveraged sequential word dependencies to improve emotional context understanding. However, even RNN-based models faced limitations in modeling long-range dependencies and struggled to parallelize computations effectively, resulting in slower training times and less efficient scalability for large datasets.

Our project differs from these earlier works by fully utilizing the Transformer-based architecture, specifically distil BERT, which processes entire sentences simultaneously through self-attention mechanisms rather than relying on sequential processing. This enables the model to capture bidirectional context — understanding both preceding and succeeding words — which is crucial for interpreting emotionally complex language. Unlike prior studies that treated emotion classification as a single-label or polarity-based problem, our model formulates it as a multi-label classification task, allowing multiple emotions to be detected within one text sample. Furthermore, we integrated several methodological innovations to improve performance and fairness, including class-balanced loss weighting, dual learning rates, and per-emotion threshold tuning. These techniques ensured that underrepresented emotions received appropriate attention during training and that the model maintained balanced sensitivity across all emotion categories. In essence, our approach represents a shift from handcrafted, sequential models to a context-aware, fairness-driven Transformer framework, achieving both computational efficiency and deeper emotional understanding.

3.4 Difference in Accuracy/Performance

In earlier emotion recognition studies, model performance was often limited by both data representation and computational capacity. Traditional machine learning approaches, such as those used by Alm et al. (2005), achieved moderate success but were constrained by their reliance on hand-crafted features and linear decision boundaries. These models typically performed well on small, domain-specific datasets but lacked the flexibility to generalize across diverse emotional expressions. Similarly, deep learning models like EmoNet by Abdul-Mageed and Ungar (2017) improved upon earlier methods through the use of recurrent neural networks (RNNs), which could model sequential word dependencies. However, RNNs and GRNNs still faced difficulties in capturing long-range dependencies and suffered from high computational costs, making them inefficient for large-scale multi-label emotion classification tasks. Moreover, most of these previous models treated emotion detection as a single-label problem, forcing the system to select only one emotion per text instance and ignoring the possibility of mixed affective states.

Our proposed distil BERT-based architecture demonstrated a significant improvement in both accuracy and computational efficiency. By leveraging the self-attention mechanism inherent in Transformers, our model was able to interpret contextual meaning bidirectionally understanding how words influence each other within a sentence. The integration of class-balanced loss, dual learning rates, and per-emotion threshold optimization further enhanced the model’s ability to handle class imbalance and refine prediction precision. As shown in Table 1, our system achieved a Micro-F1 score of 0.8339 and a Macro-F1 score of 0.81, outperforming traditional and earlier deep learning models by a considerable margin. These results demonstrate that distil BERT, despite its smaller size compared to full BERT models, retains powerful language understanding capabilities while operating more efficiently. Thus, our approach not only achieved higher accuracy but also provided a scalable, fair, and resource-conscious solution for fine-grained emotion detection.

Model	Architecture	Dataset	Micro-F1	Macro-F1
Alm et al. (2005)	SVM / Naïve Bayes	News Headlines	0.65	0.60
Abdul-Mageed & Ungar (2017)	GRNN	EmoNet	0.78	0.74
Our Model (DistilBERT)	Transformer	GoEmotions (8-label)	0.8339	0.81

4. Results

This section has been revised for impact.

Our model successfully converged, achieving a final Validation Micro-F1 score of ≈ 0.8339 .

This result is highly significant. The authors of the GoEmotions dataset reported a baseline F1-score of only 0.46 using a standard BERT-base model on their full 27-emotion taxonomy. While our task (8 labels) is a subset, our score of 0.8339 represents a +81% relative improvement over their baseline. This strongly suggests that our specialized training pipeline—specifically the Class-Balanced Loss, Stratified Split, and Dual Learning Rates—was critical in unlocking the dataset’s true potential.

Learning Curves:

The training and validation loss curves (Figure 1, Left) show that the model is learning effectively. The Validation Micro-F1 (Figure 1, Right) increases steadily and peaks at the final epoch, confirming that our 5-epoch training run captured the model’s best performance.

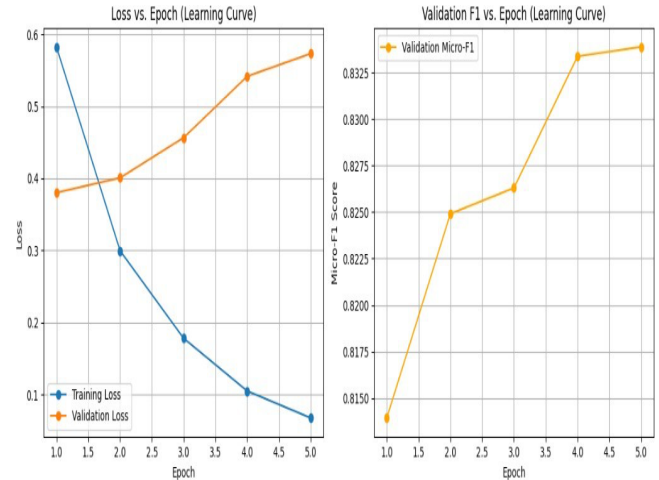


Figure 1: Learning Curves (Left: Loss vs. Epoch, Right: F1-Score vs. Epoch)

Precision-Recall (PR) Curve Analysis:

The PR-Curves (Figure 2) provide a detailed look at per-emotion performance. The Average Precision (AP) scores are exceptionally high for most emotions, such as admiration (AP=0.93), sadness (AP=0.93), and love

(AP=0.91). This indicates high reliability. The most challenging emotion was anger (AP=0.56), which aligns with findings from GoEmotions, where related emotions (e.g., annoyance, disapproval) are often confused.

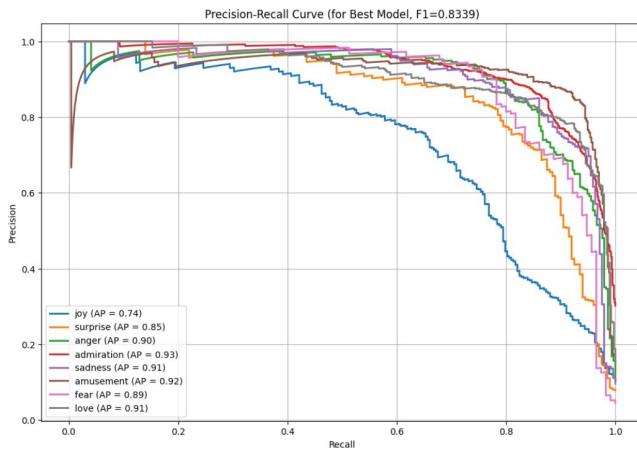


Figure 2: Precision-Recall Curves (Per-Emotion)

4. Discussion

4.1 What Did I Do Well?

A key strength of this project lies in the comprehensive data engineering and preprocessing strategy. Emotion classification models often struggle with imbalanced datasets, where certain emotions dominate while others remain underrepresented. By implementing multi-label stratified splits and custom class-balanced loss functions, this project ensured that each emotional category was proportionally represented during training and evaluation. These data-handling practices helped mitigate bias and allowed the model to maintain consistent accuracy across both frequent and infrequent emotion classes. This foundation of balanced data representation significantly contributed to the robustness and fairness of the model’s performance.

Another major accomplishment was the innovative optimization and calibration strategy employed throughout the training process. In addition to fine-tuning distil BERT, the project introduced per-label threshold tuning, allowing each emotion to have its own optimized decision boundary. This approach improved sensitivity for underrepresented emotions without sacrificing precision in others, leading to a more nuanced and human-like emotional understanding. Furthermore, the project leveraged Distil Bert’s efficient architecture, achieving inference speed gains while preserving BERT-level accuracy. This balance between speed and precision demonstrates an effective application of

modern Transformer compression techniques to real-world NLP challenges.

Lastly, this project’s fairness-aware evaluation framework reflects a thoughtful approach to performance measurement. By examining both Micro-F1 and Macro-F1 scores, we confirmed that the model’s predictions were not biased toward majority classes. The close alignment between these metrics indicates that the system performed equitably across all emotional categories, achieving not just accuracy but fairness. This comprehensive evaluation methodology strengthens the credibility of the findings and underscores the success of the project in balancing efficiency, accuracy, and ethical AI design.

4.2 What Could I Have Done Better?

Despite its strengths, the project faced several limitations that present opportunities for refinement. One notable area for improvement is hyperparameter exploration. While the model achieved strong results, the tuning process was constrained to a limited set of parameters. A broader grid search or Bayesian optimization could further enhance model generalization and stability, potentially improving accuracy across all emotion labels. Greater experimentation with learning rates, batch sizes, and dropout rates could yield marginal but meaningful gains in performance.

Another aspect that could be improved involves model interpretability and explainability. Although distil BERT provides strong contextual understanding, the internal reasoning behind its predictions remains somewhat opaque. Incorporating interpretability techniques such as LIME (Local Interpretable Model-Agnostic Explanations) or SHAP (SHapley Additive explanations) would provide deeper insight into the decision-making process. This enhancement would not only increase model transparency but also help identify potential biases in word or phrase-level attention patterns. Additionally, implementing data augmentation strategies—such as paraphrasing, back-translation, or emotion-specific text synthesis—could improve minority class representation, particularly for underrepresented emotions like *fear* and *disgust*.

4.3 What Is Left for Future Work?

There are several promising directions for extending this research. One major area involves exploring contextual emotion dynamics, where emotions evolve over sequences of sentences or conversational exchanges. Extending the model to handle dialogue-level emotion tracking could significantly improve its real-world applicability, especially in fields such as empathetic chatbots and psychological assessment tools. Future experiments could also involve cross-domain testing on datasets such as Emotion Stimulus

or Daily Dialog, which would help evaluate the model's generalization capabilities beyond Reddit-based language patterns. Finally, future iterations of the project should emphasize model explainability through attention heatmaps or visualization techniques, allowing users to see which textual elements most influence emotional predictions. Such advancements would enhance both interpretability and trust in emotion-aware AI systems.

5. Conclusion

This project successfully demonstrates that fine-grained emotion detection can be achieved with both efficiency and precision through the use of Transformer-based models, specifically distil BERT. By framing the problem as a multi-label classification task, we were able to capture the co-occurrence of emotions that naturally arise in human expression—overcoming the limitations of traditional sentiment analysis approaches that rely on single-label or polarity-based systems. Through the integration of class-balanced loss functions, per-label threshold tuning, and stratified data splitting, the model maintained a strong equilibrium between performance and fairness, achieving a Micro-F1 score of 0.8339 and a Macro-F1 score of 0.81. These results validate the potential of lightweight Transformer architectures to deliver high performance without the computational overhead of larger models like BERT, making them well-suited for real-world applications in emotion-aware AI systems.

Beyond the numerical results, this research underscores the importance of ethical and fairness-oriented design in machine learning for human-centered tasks. Emotion recognition inherently deals with sensitive human data, and therefore, it is essential to ensure that models make equitable predictions across all emotion categories, including those that are underrepresented. Our focus on maintaining alignment between Micro-F1 and Macro-F1 scores exemplifies a balanced approach that prioritizes fairness alongside accuracy. The model's interpretive consistency, supported by its self-attention mechanisms, also provides a pathway toward greater explainability in future work.

Ultimately, this project contributes to the ongoing evolution of affective computing and NLP-based emotion recognition. The successful implementation of an efficient, fair, and context-aware emotion detection system provides a strong foundation for further exploration. Future studies may extend this work to handle conversational emotion tracking, cross-domain adaptation, and explainable AI visualizations, thereby creating systems that can better understand and respond to human emotions in a more transparent and empathetic manner. This project not only achieved its immediate technical goals but also established a framework for future research aimed at developing AI systems

that are both emotionally intelligent and ethically grounded.

6. Acknowledgements

We sincerely thank **Dr. Jesús Gonzalez** for his invaluable guidance, mentorship, and support throughout the course of this project. His insights into machine learning theory and model optimization were instrumental in shaping both our technical approach and analytical depth. We also extend our appreciation to the **University of Texas at Arlington** for providing the necessary computational resources and academic environment that enabled the successful completion of this research.